

difference, known as the beat frequency, falls within the audible range of human hearing. This low-frequency beat signal is amplified by an LM358 operational amplifier, which drives a piezo buzzer.

The circuit can be assembled on a breadboard and powered by either a 9V DC power supply or a 9V battery. Fig. 1 shows the author's breadboard prototype.

Circuit and working

The circuit diagram of the BFO metal detector is shown in Fig. 2. It is built around two BC557 PNP transistors (T1 and T2), an LM358 operational amplifier (IC1), a buzzer (BZ1), and a few passive components.

The circuit consists of three main sections: two independent Colpitts radio-frequency (RF) oscillators, a passive resistive mixer, and a single-supply audio amplifier. One oscillator functions as the search oscillator, while the other serves as the reference oscillator. Under normal conditions, both oscillators operate at nearly the same frequency. When a metallic object approaches the search coil, the frequency of the search oscillator changes slightly. The outputs of the two oscillators are mixed to generate a low-frequency beat (difference) signal, which is amplified by the LM358 and applied to the buzzer. The resulting change in audio pitch indicates the presence of metal nearby. The circuit operates off a 9V battery, and capacitor C1 reduces ripple while ensuring stable operation.

The search oscillator is built around transistor T1 (BC557) and uses the custom-wound search coil L1 as its frequency-sensitive inductive element. Resistors R1 and R2 provide the required base bias, while capacitors C2 and C3, together with L1, form the Colpitts resonant tank circuit. The oscillator frequency depends on the inductance of L1 and the tank capacitance. When a metallic object comes close to the search coil, its inductance changes

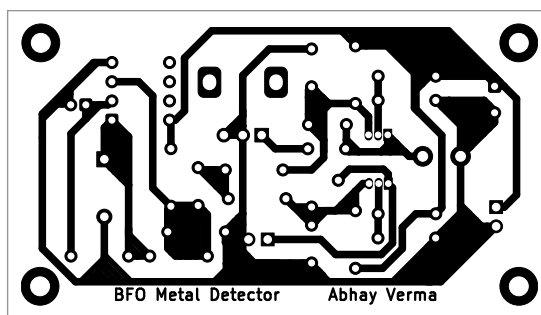


Fig. 3: Actual-size PCB

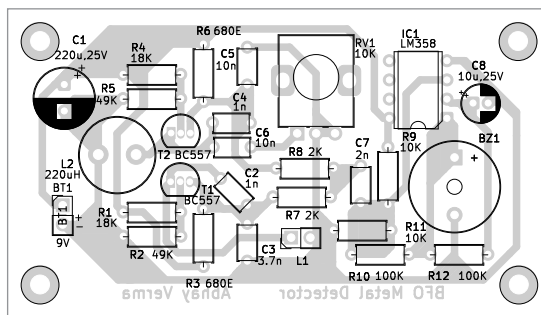


Fig. 4: Components layout

slightly, causing the oscillator frequency to shift.

The reference oscillator is built around transistor T2 (BC557) and uses the fixed inductor L2 as its resonant element. Resistors R4 and R5 provide stable base bias, while capacitors C4 and C5 form the resonant tank circuit along with L2. The oscillator is designed to operate at approximately 153kHz. Preset RV1, connected through capacitor C6, enables fine frequency adjustment, allowing the reference oscillator to be accurately tuned so that its frequency closely matches that of the search oscillator during calibration. In the absence of nearby metal, the two oscillators produce either no beat frequency or a very low audio tone.

The RF outputs from the emitters of T1 and T2 are combined at a common junction through the balanced resistive mixer formed by resistors R7 and R8. The mixing process generates the original oscillator frequencies along with their sum and difference frequencies. Capacitor C7, together with the mixer network, acts as a low-pass

filter that suppresses the high-frequency RF components, leaving only the low-frequency beat signal. This beat signal is then AC-coupled to the amplifier stage through capacitor C7, which blocks the DC component.

The beat signal is applied to the non-inverting input (pin 3) of the LM358 operational amplifier (IC1A). Since the circuit uses a single 9V supply, resistors R9 and R11 generate a virtual ground of approximately 4.5V to bias the operational amplifier correctly. Resistors R10 and R12, together with capacitor C8, form the feedback network and configure

the amplifier with a voltage gain of approximately 100, enabling amplification of the weak beat signal.

The operation of the circuit is straightforward. When a metallic object is brought near the search coil, its inductance changes slightly, causing the oscillator frequency to shift. This changes the beat frequency, producing a corresponding change in the buzzer tone that indicates the presence of metal.

Construction and testing

The actual-size, single-sided PCB layout of the BFO metal detector is shown in Fig. 3, and its component layout is shown in Fig. 4. If the designed PCB is unavailable, the circuit can be assembled on a good-quality general-purpose PCB. The author's prototype was assembled on a breadboard, as shown in Fig. 1.

During assembly, keep all RF signal connections as short as possible to minimise parasitic capacitance and high-frequency noise. Place the custom search coil (L1) away from metal workbenches or other metallic objects during testing. Position the decoupling capacitor